

# Yasra Chandio

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## Education

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|------------------------------------------------------------------------------------------------------|-----------|
| PhD in Computer Engineering, University of Massachusetts Amherst.                                    | 2020–2025 |
| MS in Computer Science, Binghamton University (SUNY).                                                | 2018–2019 |
| BS in Software Engineering, Mehran University of Engineering and Technology, Pakistan [Bronze Medal] | 2010–2014 |

## Awards Highlights

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| MIT EECS Rising Star   19% acceptance rate   <a href="#">Link</a><br><i>A highly selective workshop based on academic excellence and commitment to advancing equity and inclusion.</i>                                        | 2024 |
| Cyber-Physical Systems (CPS) Rising Star   16% acceptance rate   <a href="#">Link</a><br><i>A highly selective workshop based on outstanding achievements and future potential in CPS and related areas.</i>                  | 2024 |
| UMASS College of Engineering Diversity, Equity, and Inclusion (DEI) Award   <a href="#">Link</a><br><i>Award recognizing one undergraduate/graduate student/postdoc recognizing exceptional contributions to DEI.</i>         | 2024 |
| Research featured at national and international news outlets   <a href="#">BBC</a> , <a href="#">UMASS</a> , <a href="#">CoE</a> , <a href="#">ScienceDaily</a> , <a href="#">TechXplore</a>                                  | 2024 |
| Young Researcher at 10th Heidelberg Laureate Forum (HLF)   11% acceptance rate   <a href="#">Link</a><br><i>A prestigious international forum inviting 200 exceptional young researchers in mathematics and CS worldwide.</i> | 2023 |
| Best Ph.D. Forum Presentation Award at ACM SenSys                                                                                                                                                                             | 2022 |
| Computing Research Association-Education (CRA-E) Graduate Fellow   Two-year term   <a href="#">Link</a><br>National computer research committee offers a graduate student role advocating undergraduate research.             | 2022 |
| Three Minute Thesis (3MT) finalist at UMASS   <a href="#">Talk</a> <a href="#">Link</a>                                                                                                                                       | 2022 |

## Publications

My work has been published at the top venues in virtual, augmented, and mixed reality (VR/AR/MR), robotics, systems, and machine learning, including IEEE VR, TVCG, IEEE/RSJ IROS, NeurIPS, ACM FAccT, and BuildSys/e-Energy.

- [1] Yasra Chandio, Momin Ahmed Khan, Khotso Selialia, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar. [A Neurosymbolic Approach To Adaptive Feature Extraction In SLAM](#). In: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). 2024.
- [2] Yasra Chandio, Victoria Interrante, and Fatima M. Anwar. [Balancing Presence And Safety Using Reaction Time In Mixed Reality](#). In: Workshop at IEEE International Symposium on Mixed and Augmented Reality (SafeAR@ISMAR). 2024.
- [3] Yasra Chandio, Victoria Interrante, and Fatima M. Anwar. [Human Factors At Play: Understanding The Impact Of Conditioning On Presence And Reaction Time In Mixed Reality](#). In: IEEE Conference on Virtual Reality and 3D User Interfaces (VR). 2024. [selected to appear in IEEE TVCG \[Top 12% paper\]](#).
- [4] Yasra Chandio, Noman Bashir, Tian Guo, Elsa Olivetti, and Fatima M. Anwar. [Scoping The Decarbonization Of Collaborative Mixed Reality](#). In: IEEE International Symposium on Emerging Metaverse (ISEMV). 2024.
- [5] Yasra Chandio, Noman Bashir, and Fatima M. Anwar. [Stealthy and practical multi-modal attacks on mixed reality tracking](#). In: IEEE International Conference on Artificial Intelligence & extended and Virtual Reality (AIxVR). 2024. [Best Paper Finalist](#).
- [6] Khotso Selialia, Yasra Chandio, and Fatima Anwar. [Mitigating feature heterogeneity biases in Federated Learning](#). In: ACM Conference on Fairness, Accountability, and Transparency (FAccT). 2024.
- [7] Momin Ahmed Khan, Yasra Chandio, and Fatima Anwar. [HYDRA-FL: Hybrid Knowledge Distillation for Robust and Accurate Federated Learning](#). In: Annual Conference on Neural Information Processing Systems (NeurIPS). 2024.
- [8] Yasra Chandio, Noman Bashir, Victoria Interrante, and Fatima M. Anwar. [Investigating the Correlation Between Presence and Reaction Time in Mixed Reality](#). In: IEEE Transactions on Visualization and Computer Graphics (TVCG) (2023).

- [9] Noman Bashir, **Yasra Chandio**, David Irwin, Fatima Anwar, Jeremy Gummesson, and Prashant Shenoy. *Jointly managing electrical and thermal energy in solar- and battery-powered computer systems*. In: ACM International Conference on Future Energy Systems (e-Energy). 2023.
- [10] **Yasra Chandio**, Noman Bashir, and Fatima M. Anwar. *Dataset: HoloSet - A dataset for visual inertial pose tracking in extended reality*. In: 5th International SenSys/BuildSys Workshop on Data (DATA). 2022.
- [11] **Yasra Chandio**. *Towards secure and immersive mixed reality: PhD Forum Abstract*. In: 20th ACM Conference on Embedded Networked Sensor Systems (SenSys). 2022. **Best Presentation Award**.
- [12] Khotso Selialia, **Yasra Chandio**, and Fatima M. Anwar. *Federated Learning Biases in Heterogeneous Edge-Devices - A Case-Study*. In: 4th International SenSys/BuildSys Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengeloT). 2022.
- [13] **Yasra Chandio** and Fatima M. Anwar. *Poster: Spatiotemporal Security in Mixed Reality systems*. In: 18th ACM Conference on Embedded Networked Sensor Systems (SenSys). 2020.
- [14] **Yasra Chandio**, Aditya Mishra, and Anand Seetharam. *GridPeaks: Employing Distributed Energy Storage For Grid peak Reduction*. In: IEEE International Green and Sustainable Computing conference (IGSC). 2019.
- [15] **Yasra Chandio**, Jo A gila Bitsch, Affan A. Syed, and Muhammad Hamad Alizai. *Networking Wireless Energy in Embedded Networks*. In: ACM Transactions on Sensor Networks (TOSN) (2018).
- [16] Samar Abbas, Abu Bakar, **Yasra Chandio**, Khadija Hafeez, Ayesha Ali, Tariq M Jadoon, and Muhammad Hamad Alizai. *Inverted HVAC: Greenifying Older Buildings, One Room at a Time*. In: ACM Transactions on Sensor Networks (TOSN) (2018).
- [17] Khadija Hafeez, **Yasra Chandio**, Abu Bakar, Ayesha Ali, Affan A. Syed, Tariq M. Jadoon, and Muhammad Hamad Alizai. *Inverting HVAC for Energy Efficient Thermal Comfort in Populous Emerging Countries*. In: ACM International Conference on Systems for Energy-Efficient Built Environments (BuildSys). 2017. **Audience Choice Award**.
- [18] M. Hamad Alizai, Qasim Raza, **Yasra Chandio**, Affan A. Syed, and Tariq M. Jadoon. *Simulating Intermittently Powered Embedded Networks*. In: 13th International Conference on Embedded Wireless Systems and Networks (EWSN). 2016.

#### UNDER-REVIEW/IN-PREPARATION

- [1] **Yasra Chandio**, Khotso Selialia, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar. *Lost in Tracking Translation: A Comprehensive Analysis of Visual SLAM in Human-Centered XR and IoT Ecosystems*. **[under review]**.
- [2] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar. *Reaction Time as a Proxy for Presence in Mixed Reality with Distraction*. **[under review]**.
- [3] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar. *Tap into Reality: Understanding the Impact of Interactions on Presence and Reaction Time in Mixed Reality*. **[under review]**.
- [4] **Yasra Chandio**, Victoria Interrante, and Fatima M. Anwar. *Towards Objective And Implicit Presence Measures In Mixed Reality*. **[Short paper][under review]**.
- [5] **Yasra Chandio**, Luis Antonio Garcia, Joseph DeGol, and Fatima M. Anwar. *ContinuumTrack: Adaptive Real-Time Profiling and Correction in Dynamic SLAM Systems*. **[in preparation]**.
- [6] Khotso Selialia, **Yasra Chandio**, Jimi Oke, and Fatima Anwar. *LipFed: Mitigating Subgroup Bias in Federated Learning with Lipschitz Constraints*. **[under review]**.
- [7] Khotso Selialia, **Yasra Chandio**, Jimi Oke, and Fatima Anwar. *Tackling Group Fairness Inconsistencies in Federated Learning*. **[in preparation]**.
- [8] Diana Romero\*, **Yasra Chandio\***, Fatima Anwar, and Salma Elmalaki. *Sensing Group Behavior In Mixed Reality*. **[in preparation]**[\* these authors contributed equally.]
- [9] John Murray, **Yasra Chandio**, Fatima Anwar, and Michael Zink. *Exploring Relationship Between Quality Of Experience And Quality Of Service In Collaborative Mixed Reality*. **[in preparation]**.
- [10] Momin Ahmad Khan, Virat Shejwalkar, **Yasra Chandio**, Amir Houmansadr, and Fatima M. Anwar. *A Critical Analysis Of The Pitfalls In Federated Learning Defense Evaluations*. **[under review]**.
- [11] Abdullah Al Ishtiaq, Raja Hasnain Anwar, **Yasra Chandio**, Fatima M. Anwar, Syed Rafiul Hussain, and Muhammad Taqi Raza. *Cloud Nine Connectivity: Evading in-Flight Paywalls For Free Internet*. **[under review]**.

## Awards and Honors

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|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| UMASS Campus Climate Improvement Grant   \$2500<br><i>Support projects that foster connection, build community, and promote a more inclusive campus.</i> | 2024  |
| Scholar at CRA-WP Grad Cohort Workshop for Inclusion, Diversity, Equity, Accessibility, and Leadership Skills                                            | 2024  |
| IEEE Robotics and Automation Society (RAS) travel grant to attend IROS                                                                                   | 2024  |
| Abbe Grant to attend HLF   Highly selective grant, given to 30/200 HLF attendees                                                                         | 2023  |
| Inaugural student inductee in UMASS ECE DEI Committee                                                                                                    | 2024  |
| CPS-IoT Week SIGBED student travel award   declined                                                                                                      | 2023  |
| Scholar at CRA-WP Grad Cohort                                                                                                                            | 2022  |
| Invited to attend Google PhD Fellowship Summit                                                                                                           | 2021. |
| Honorable mention in UMASS ECE 3MT                                                                                                                       | 2021  |
| Selected for Computer Research Mentorship (CSRMP) Program at Google                                                                                      | 2021  |
| Grace Hopper Scholar to attend Grace Hopper Celebration                                                                                                  | 2019  |
| National Science Foundation travel grant to attend IEEE IGSC                                                                                             | 2019  |
| Summer Provost Fellowship at Binghamton University                                                                                                       | 2019  |
| Best Paper Audience Choice Award at ACM BuildSys                                                                                                         | 2017  |
| ACM SenSys Travel grant to attend ACM BuildSys/SenSys                                                                                                    | 2017  |

## Media Coverage

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|--------------------------------------------------------------------------------------------------------------------|--------------|
| <a href="#">Can AR and VR finally disrupt the exhausting culture of video meetings?</a>                            | BBC          |
| <a href="#">ECE's Yasra Chandio Qualifies as a "Rising Star" in Both Actual and Virtual Reality</a>                | CoE          |
| <a href="#">College of Engineering Selects LaChance and Chandio for Its Diversity, Equity, and Inclusion Award</a> | CoE          |
| <a href="#">CRA Education Committee Selects New Graduate Fellow</a>                                                | CRA          |
| <a href="#">New Study: Immersive Engagement In Mixed Reality Can Be Measured With Reaction Time</a>                | UMASS        |
| <a href="#">Immersive Engagement in Mixed Reality Can Be Measured with Reaction Time</a>                           | ScienceDaily |
| <a href="#">Study: Immersive Engagement in Mixed Reality Can Be Measured with Reaction Time</a>                    | TechXplore   |

## Employment

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|-----------------------------------------------------------------------------------------------|-----------|
| Graduate Research Assistant, Emerging Technologies Lab, UMASS Amherst   Advisor: Fatima Anwar | 2020–on   |
| Graduate Research Assistant, CS Department, Binghamton University   Advisor: Gaunhua Yan      | 2018–2019 |
| Research Associate, Systems and Networks Lab, LUMS, Pakistan   Advisor: Muhammad Hamad Alizai | 2015–2017 |
| Research Engineer, NUCES, Pakistan   Advisor: Affan A. Syed                                   | 2014–2015 |

## Teaching and Mentorship

### Teaching Assistant

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|-------------------------------------------------------------------------------------------------|--------------|
| ECE 524/624: Advanced Embedded Systems Design (Graduate) – <b>helped design course projects</b> | Fall 21–23   |
| ECE 322: Systems Programming                                                                    | Fall 24      |
| ECE 231: Introduction to Embedded System – <b>helped design new labs</b>                        | Spring 21–24 |
| ECE 571 (SUNY): Programming Languages (Graduate)                                                | Fall 19      |
| CS 677 (LUMS): Internet of Things                                                               | Fall 16      |

### Mentorship

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|----------------------------------------------------------------------------------------------------------------|-------|
| <u>John Murray</u> (UMASS PhD)                                                                                 | 2024– |
| <u>Diana Romero</u> (UC Irvine PhD)                                                                            | 2023– |
| <u>Standly Laguerre</u>   REU at NSF Computing for an Equitable Energy Transition program (Macalester College) | 2023  |

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| –Project: <i>Unpacking Visual SLAM: A Critical Analysis of Tracking Methodologies and Their Pitfalls</i> |      |
| <u>Kenneth Lin, Yav Rohatgi (UMASS BS)</u>                                                               | 2023 |
| –Project: <i>Remote Interactions in Mixed Reality</i>                                                    |      |
| <u>Colin Leydon, Eyado Ocho (UMASS BS)</u>                                                               | 2023 |
| –Project: <i>Deep Learning Bases Tracking in Mobile Tracking Systems</i>                                 |      |
| <u>John Dale, Dani Kasti (UMASS BS)</u>                                                                  | 2022 |
| –Project: <i>Objectively Measuring Multi-User Collaboration in Mixed Reality</i>                         |      |
| <u>Yifan Meng, Linghao Meng, Shaoyu Dai, Yanan Zhang (UMASS MS)</u>                                      | 2022 |
| –Project: <i>Pose Estimation for Micro Movements in Mixed Reality</i>                                    |      |
| <u>John Murray, Zachary Wannie, Alexander Dickopf, Zhehang Zhang (UMASS MS)</u>                          | 2021 |
| –Project: <i>Deep Learning Based Sensor Fusion for Mixed Reality systems</i>                             |      |
| <u>Subhankar Chowdhury, Abhinaba Das (UMASS MS)</u>                                                      | 2021 |
| –Project: <i>Deep Learning Based Sensor Fusion for IoT systems</i>                                       |      |
| <u>Xiaohao Xia, Yinxuan Wu, Xintao Ding, Xin Zhao (UMASS MS)</u>                                         | 2021 |
| –Project: <i>Remote collaboration in Virtual Reality</i>                                                 |      |
| <u>Nikhil Sarecha, Rehmat Singh Kang (UMASS MS)</u>                                                      | 2020 |
| –Project: <i>Remote collaboration in Mixed Reality with IoT devices</i>                                  |      |

### Guest Lectures

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|------------------------------------------------------------------------------|------|
| Deep Learning based Tracking for Mobile Systems (UMASS ECE 524/624)          | 2023 |
| Mixed Reality Security and Privacy (UMASS ECE 524/624)                       | 2022 |
| Introduction to Embedded System with <i>BeagleBone Black</i> (UMASS ECE 231) | 2020 |

## Service

### Service to the Profession

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|---------------------------------------------------------------------------------------------------|------|
| Reviewer at IEEE Conference on Virtual Reality And 3D User Interfaces ( <a href="#">IEEE VR</a> ) | 2025 |
| Reviewer at ACM Conference on Human Factors in Computing Systems ( <a href="#">ACM CHI</a> )      | 2025 |
| Reviewer at IEEE International Conference on Robotics & Automation ( <a href="#">IEEE ICRA</a> )  | 2024 |
| Reviewer at International Conference on Learning Representations ( <a href="#">ICLR</a> )         | 2024 |
| Technical Program Committee member at <a href="#">DATA@Buildsys</a>                               | 2024 |
| Technical Program Committee member at <a href="#">ACM S3@Mobicom</a>                              | 2024 |
| Student Volunteer at ACM SenSys                                                                   | 2022 |
| Shadow Program Committee member at ACM SenSys                                                     | 2022 |
| Shadow Program Committee member at ACM EuroSys                                                    | 2022 |
| Review Committee member at Extended Reality track at Grace Hopper Celebration (GHC)               | 2022 |
| Review Committee member at VR/AR/MR track at Grace Hopper Celebration (GHC)                       | 2021 |

### Service to the Community

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| <u>Chair</u> College of Engineering Dean's Graduate Advisory Group                                                                          | 2024–2025 |
| – Students from the entire college are selected to be part of the group, and then the group votes to elect the Chair.                       |           |
| <u>Organizer</u> UMASS ECE Women Dinner Series   <a href="#">[Link]</a>                                                                     | 2024–2025 |
| – Raised \$2500 with a competitive UMASS Campus Climate Grant for community building through a series of dinners with workshops and panels. |           |
| <u>Graduate Student Member</u> of ECE DEI Committee                                                                                         | 2023–2025 |
| <u>Founding Chair</u> , ECE Grad Gender Diversity group                                                                                     | 2023–2025 |
| <u>Organizer</u> , ECE grad women retreats   Raised \$1700                                                                                  | 2023–2025 |

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| <u>Organizer and Panelist</u> , ECE Graduate Orientation                                                                                                                                 | 2023–2025 |
| <u>Panel Organizer &amp; Moderator</u> , UMASS Women in Engineering Day                                                                                                                  | 2023–2025 |
| –A panel discussion connecting industry professionals with high school students to discuss careers in engineering.                                                                       |           |
| <u>Editor</u> CRA-Undergraduate Research Highlights   <a href="#">[Link]</a>                                                                                                             | 2022–2024 |
| – The series features outstanding undergraduate research across North America, offering insights into research experiences, mentorship, career advice, and promoting graduate education. |           |
| <u>Instructor and Contributor</u> , UMass Turing Summer Program   <a href="#">[Link]</a>                                                                                                 | 2022–2025 |
| <u>Panelist</u> at Graduate Women In STEM (GWIS) Undergraduate Panel                                                                                                                     | 2024      |
| – A panel discussion connecting UMASS graduate and undergraduate students to discuss careers in STEM.                                                                                    |           |
| <u>Instructor</u> , UMass K-12 Engineering Outreach Summer Program                                                                                                                       | 2021      |
| <u>Presenter</u> , Pioneer Valley Chinese Immersion Charter School                                                                                                                       | 2024      |
| <u>Volunteer</u> , Amherst Survival Center                                                                                                                                               | 2020      |
| <u>Volunteer</u> , Binghamton University Commencement                                                                                                                                    | 2019      |

## References

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|----------------------------------------------------------------------------------------------|-------------------------|
| 1. <a href="#">Fatima Anwar</a> , Assistant Professor at University of Massachusetts Amherst | fanwar@umass.edu        |
| 2. <a href="#">Victoria Interrante</a> , Professor at University of Minnesota, Twin Cities   | interran@umn.edu        |
| 3. <a href="#">Michael Zink</a> , Professor at University of Massachusetts Amherst           | mzink@umass.edu         |
| 4. <a href="#">Luis Antonio Garcia</a> , Assistant Professor at University of Utah           | la.garcia@utah.edu      |
| 5. <a href="#">Joseph DeGol</a> , CTO at Steg AI                                             | jomnipotent17@gmail.com |